

TaiNing (Doris) Wang
Assistant Professor, Department of Computer Science
Loyola University Chicago
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Professional Experience

- **Tenure-Track Assistant Professor**, Department of Computer Science
Loyola University Chicago, United States Aug 2023 – Present
- **Postdoctoral Research Fellow**
National University of Singapore, Singapore Apr 2022 – Apr 2023
Supervisor: Prof. Kian-Lee Tan
- **Research Intern**
Microsoft Research, Redmond, United States May 2018 – Aug 2018
Supervisor: Dr. Vivek Narasayya
- **Research Intern**
INRIA (Computer Science Laboratory), École Polytechnique, France May 2015 – Jul 2015
Supervisor: Prof. Mireille Régnier
Project: Efficient representation of highly similar strings
- **Research Intern**
Agency for Science, Technology and Research (A*STAR), Singapore May 2014 – Aug 2014
Data Storage Institute
Supervisor: Dr. Shuqin Ren
Project: Search and computation over cloud-based encrypted document store

Education

- Ph.D. in Computing (Computer Science) Aug 2016 – May 2022
National University of Singapore
Advisor: Prof. Chee-Yong Chan
Thesis: *Join Query Optimization with Improved Selectivity Estimation and Complete Join Reorderability*
- Bachelor of Computing (Computer Science), Honors (Highest Distinction) Aug 2012 – Jun 2016
National University of Singapore
GPA: 4.84 / 5.00 (Second best student in the cohort)
- Undergraduate Exchange Program Jan 2015 – May 2015
University of North Carolina at Chapel Hill

Research Interests

Database systems, query processing and optimization, graph data, applied AI and ML, AI accountability, formal methods, data sampling

Publications

Refereed Conference Publication

- Mohammed Abuhamad, Brian Ortiz, Arslan Bisharat, Eric Spencer, Khushboo Bhadauria, TaiNing Wang, George Thiruvathukal, Konstantin Laufer. **Can LLMs Write Correct TLA+ Specifications? Evaluating Natural-Language-to-TLA+ Generation.** *International Conference on Software Technologies (ICSOFT)*, 2026. Accepted.
- TaiNing Wang, Yunpeng Niu, Chee-Yong Chan. **Complete Join Reordering for Null-Intolerant Joins.** *IEEE International Conference on Data Engineering (ICDE)*, 2023.
- TaiNing Wang, Chee-Yong Chan. **Improved Correlated Sampling for Join Size Estimation.** *IEEE International Conference on Data Engineering (ICDE)*, 2020.
- TaiNing Wang, Chee-Yong Chan. **Improving Join Reorderability with Compensation Operators.** *ACM International Conference on Management of Data (SIGMOD)*, 2018.
- Shuqin Ren, Benjamin Hong Meng Tany, Sivaraman Sundaram, TaiNing Wang, Khin Mi Mi Aung. **Homomorphic Exclusive-Or Operation Enhance Secure Searching on Cloud Storage.** *IEEE International Conference on Cloud Computing Technology and Science (IEEE CloudCom)*, 2014.

Refereed Journal Publication

- Shuqin Ren, Benjamin Hong Meng Tany, Sivaraman Sundaram, TaiNing Wang, Yibin Ng, Victor Chang, Khin Mi Mi Aung. **Secure Searching on Cloud Storage Enhanced by Homomorphic Indexing.** *Future Generation Computer Systems*, 2016.

Submitted and Under Review

- Amy Kuhl, TaiNing Wang, Yas Silva. **Structural Clustering on Uncertain Weighted Graphs.** Submitted, 2026.

Technical Report

- TaiNing Wang, Yunpeng Niu, Chee-Yong Chan. **Extended Technical Report on Complete Join Reordering for Null-Intolerant Joins.** Technical Report, 2023.

Teaching

Loyola University Chicago

- COMP 163: Discrete Structures
Course Instructor Fall 2023 – Present

National University of Singapore

- CS2102: Database Systems
Teaching Assistant 2018 – 2020
- CS4224: Distributed Databases
Teaching Assistant 2018 – 2020

Service

Program Committee

- SIAM International Conference on Data Mining (SDM), 2026

Reviewer

- SIAM International Conference on Data Mining (SDM), 2026
- IEEE Transactions on Cloud Computing (IEEE TCC), 2024
- ACM Transactions on Database Systems (ACM TODS), 2020

Sub-reviewer

- IEEE International Conference on Data Engineering (IEEE ICDE), 2021, 2020, 2018
- International Conference on Very Large Data Bases (VLDB), 2019, 2018

Organizing Committee

- The Greater Chicago Area Systems Research Workshop (GCASR), Poster Co-Chair, 2026

Honors and Awards

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| • Microsoft Research Asia Fellowship | 2019 |
| • President Graduate Fellowship, National University of Singapore | 2018 – 2020 |
| • Research Achievement Award, National University of Singapore | 2018 |
| • Tata Consultancy Services Asia Pacific Medal & Prize | 2016 |
| • Science & Technology Undergraduate Scholarship, Singapore | 2012 – 2016 |
| • Dean's List Award, National University of Singapore | 2013 – 2014 |
| • Dean's List Award, University of North Carolina at Chapel Hill | 2015 |

Invited Talks

Complete Join Reordering for Null-Intolerant Joins

- IEEE ICDE conference 2023

Improved Correlated Sampling for Join Size Estimation

- IEEE ICDE conference 2020

Improving Join Reorderability with Compensation Operators

- Microsoft Research Asia, 2019
- SoC Research Week, National University of Singapore, 2019
- ACM SIGMOD conference 2018
- Microsoft Research Redmond, 2018